Persistent Narrowband Radio Lines Below the Ionospheric Cutoff: 27 Years of Wind and Five Years of STEREO

Charles C. Norton

Abstract

Below about 10 MHz the ionosphere reflects radio waves, so a transmitter at those frequencies can be heard only from space. We search every one-minute spectrum of the Wind/WAVES receivers from 1995 to 2021, 20 kHz to 13.8 MHz, and every spectrum of STEREO-A in 2008, 2012 and 2020 and of STEREO-B in 2010 and 2012, 125 kHz to 16 MHz, for lines that persist: a channel five robust deviations above its neighbours in more than one per cent of the minutes on thirty days or more. Every persistent line has a known source. The solar wind's plasma line, Earth's auroral kilometric radiation and the terrestrial broadcast, maritime, aeronautical and amateur services account for most, and terrestrial leakage is recognised by its day-to-day correlation with the broadcast lines of the 41 m band, which reaches the spacecraft only through a sunlit ionosphere. The rest are made on the spacecraft: Wind's lines at 8025 and 1175 kHz, which do not follow the broadcast lines, and lines that one STEREO spacecraft holds five times as often as Wind or the other spacecraft, among them 10425 kHz in both STEREO receivers, whose occupancy in STEREO-A rose from 0.08 per cent in 2008 to 4.7 per cent in 2020. No persistent line remains outside the known sources in any of the 32 spacecraft-years. The channels are 3, 20 and 50 kHz wide, so the search is for narrow lines only in the sense that a line fills one channel; a 5σ line in Wind's RAD2 carries $2 \times 10^{-16}\,\mathrm{W\,m^{-2}}$, the output of a 6×10^{7} W isotropic source at 1 au or of 2×10^{18} W at 1 pc, so the search is one of the Solar System and of the strongest conceivable transmitters at the nearest stars; Wind's RAD1 measures only 32 to 38 of its 256 channels directly, which alone carry its search, and 64 per cent of RAD2's band and 52 per cent of STEREO's lie inside the terrestrial service bands, where a line is assigned to the service and not counted.

1 Introduction

Radio searches for technosignatures listen from the ground, where the ionosphere reflects everything below its plasma frequency of a few to about ten megahertz, and searches at low frequencies, such as one with the Murchison Widefield Array near 100 MHz [1], observe well above it. The band from tens of kilohertz to ten megahertz is open only to receivers in space. The interstellar plasma frequency is a few kilohertz, but free-free absorption [2] in the warm, partly ionised clouds around the Sun, with $n_e \approx 0.1\,\mathrm{cm^{-3}}$ at about 7000 K [3], sets the reach: across 5 to 10 pc of such gas the optical depth is about 0.1 at 1 MHz, 1 near 300 kHz and 10 at 100 kHz, so nearby stars are heard through it above a few hundred kilohertz, depending on the sightline, and below about 100 kHz only sources within the heliosphere or across the shortest paths through warm gas. Spacecraft built to study the solar wind and the magnetosphere have recorded that band continuously for three decades: the WAVES receivers of Wind [4], since 2004 at the Sun-Earth L1 point, and those of the twin STEREO spacecraft [5], which drifted ahead of and behind Earth on its orbit from 2006.

Their spectra are crowded with natural lines and with terrestrial ones. Earth's auroral kilometric radiation reaches far beyond the magnetosphere [6], and terrestrial transmitters leak through the

ionosphere where it is sunlit, as the Radio Astronomy Explorer found [7]. A narrowband transmitter elsewhere would add a line that persists in one channel, whose behaviour from minute to minute and day to day separates it from these. This paper searches the full Wind and STEREO records for such lines.

1.1 Contributions

1. A search of 27 years of Wind/WAVES and 5 years of STEREO/WAVES spectra for persistent narrowband lines below the ionospheric cutoff.
2. Two tests that separate terrestrial and spacecraft lines from the rest: a line's day-to-day correlation with the broadcast lines, and its occupancy in the other spacecraft, which share the sky but not the receiver.

2 Data and method

Data. Wind's RAD1 receiver covers 20 to 1040 kHz in 256 channels 4 kHz apart and 3 kHz wide, and RAD2 1.075 to 13.825 MHz in 256 channels 50 kHz apart and 20 kHz wide [4]; CDAWeb's WI_H1_WAV files give one-minute averages of both for every day from 1995 to 2021, and no later year is in the archive. RAD2 measures every channel. RAD1 measures 32 to 38 of its channels in nearly every minute of every year, from 20 to 1040 kHz at spacings that grow from 4 kHz at the bottom to about 100 kHz at the top, and the archive fills the other channels by interpolation between the measured ones, which the files mark through each day's minimum voltage. STEREO's high-frequency receiver covers 125 kHz to 16 MHz in 50 kHz channels about every 39 s, from CDAWeb's level-3 files, whose spectra are calibrated in $\mathrm{W\,m^{-2}\,Hz^{-1}}$ after the subtraction of a daily background. We take every spectrum of STEREO-A in 2008, 2012 and 2020 and of STEREO-B in 2010 and 2012, years spread over the spacecraft's drift from Earth and, for 2012, held by both.

Lines. In each spectrum a channel carries a line when it stands five robust deviations above the median of its eight neighbouring channels. A channel's occupancy is the fraction of a year's spectra in which it carries a line, and a line is persistent when its occupancy exceeds one per cent on thirty days or more. In RAD1 a measured channel stands above interpolated neighbours whenever the spectrum curves, so its lines are tested among the measured channels alone, each against the median of its eight nearest measured neighbours. A line narrower than a channel is diluted over the channel's width, so the statistic is a line's power against the background in the channel, and a line is narrowband here only in the sense of filling one channel of 3, 20 or 50 kHz.

Known sources. A persistent line is assigned, in order, to the spacecraft's own lines; to the shortwave broadcast bands within 25 kHz; to Earth's auroral kilometric radiation between 100 and 700 kHz and the solar wind's plasma line below 100 kHz; to the maritime and aeronautical mobile bands within 25 kHz and the amateur bands; to broadcasting up to 200 kHz outside the broadcast bands; and to the terrestrial sources whose occupancy by hour of UT varies by more than its mean, since Earth's transmitters wax and wane as it turns. The service bands with their 25 kHz margins cover 64 per cent of RAD2's range and 52 per cent of STEREO's, and with the 200 kHz of out-of-band broadcasting 76 and 62 per cent; the auroral and plasma-line ranges cover 67 per cent of RAD1's. A line inside those ranges is assigned to the service or the natural source and not counted, so the search for lines outside the known sources covers the remaining quarter of RAD2's band, 38 per cent of STEREO's, and the third of RAD1's above 700 kHz.

Sensitivity. The channel-to-channel scatter of a one-minute spectrum sets what a line must carry. In RAD2 the robust deviation of a channel from the median of its eight neighbours is 2.4 per

cent of the background power, in every year sampled (1999 and 2020), so a 5σ line adds 12 per cent of the background to a channel. Above 1 MHz the background is the Galaxy's, whose brightness is close to $10^{-20}\,\mathrm{W\,m^{-2}\,Hz^{-1}\,sr^{-1}}$ from 2 to 10 MHz [9], and a short dipole collects it from $8\pi/3$ sr, so the background's equivalent flux density is $8 \times 10^{-20}\,\mathrm{W\,m^{-2}\,Hz^{-1}}$ and the threshold for a line filling RAD2's 20 kHz is $F = 5 \times 0.024 \times 8 \times 10^{-20} \times 2 \times 10^4 \approx 2 \times 10^{-16}\,\mathrm{W\,m^{-2}}$, for a source at the antenna's best direction. That is the flux of an isotropic source of 6×10^7 W at 1 au, 2×10^{18} W at 1 pc and 2×10^{20} W at 10 pc, or of a transmitter of 5×10^7 W behind a kilometre-wide aperture at 1 pc, whose beam at 5 MHz is 60 rad across and gains nothing. STEREO's calibrated spectra scatter by 6 to 7 per cent of a background of $1.4 \times 10^{-19}\,\mathrm{W\,m^{-2}\,Hz^{-1}}$ above 2 MHz in the raw receiver power and by 23 per cent of the background-subtracted level, so its 50 kHz channels reach about $2 \times 10^{-15}\,\mathrm{W\,m^{-2}}$. RAD1's measured channels scatter about the median of their measured neighbours by 3 per cent of the background in 2020 and by 10 per cent above 300 kHz and 29 per cent below it in 1999, where the auroral emission varies from minute to minute, so a line must exceed its neighbours by 15 per cent of the background in the one year and by half to one and a half times it in the other; with the galactic brightness absorbed to a few $10^{-21}\,\mathrm{W\,m^{-2}\,Hz^{-1}}$ below 1 MHz the threshold in a 3 kHz channel is of order 10^{-17} to $10^{-16}\,\mathrm{W\,m^{-2}}$, where the receiver's own noise, which dominates the lowest channels, raises it further.

Day-to-day correlation. Terrestrial leakage reaches the spacecraft only through a sunlit ionosphere, so every leaked line rises and falls from day to day with the broadcast lines of the 41 m band (7225 to 7375 kHz). The Spearman correlation of a line's daily occupancy with theirs separates leakage from a line made on the spacecraft or arriving from beyond Earth.

Spacecraft lines. A narrowband source beyond the spacecraft reaches Wind and both STEREO spacecraft, whose receivers are about equally sensitive, about equally often. A line that one spacecraft holds five times as often as Wind, or as the other STEREO spacecraft in the same year, is made on that spacecraft.

Instrument changes. Two intervals change Wind's spectra. During its spacecraft anomaly of 2014 October 26 to 2015 January 30 the 1175 kHz channel switched on and off, and in 2014 and 2015 RAD1's list of measured channels shrank to 15; both are set aside.

3 Results

Wind. In every year from 1995 to 2021 the persistent lines all have known sources. In RAD2 there are 119 persistent lines in 1995, 153 in 1996, 112 in 1998 and 153 in 1999: in 1999, 55 in the broadcast bands, 36 in the maritime and 13 in the aeronautical bands, 14 in out-of-band broadcasting, 2 in the amateur bands, 31 with the diurnal modulation of a terrestrial source, and 2 of Wind's own. One line outside the broadcast bands, 6975 kHz in 1997, follows the 41 m lines from day to day (Spearman 0.52), as terrestrial leakage must, and so does 7175 kHz in 2020 (0.53). The steady line at 8025 kHz, present on 365 and 330 days of 2020 and 2021 with no daily modulation, does not (-0.16, against 0.74 for the broadcast line at 9625 kHz), so it is made on Wind; so are the lines at 1175 kHz and, in 1995, at 1275 and 1525 kHz. In RAD1 the test against all neighbouring channels marks 55 to 96 lines a year, every one a measured channel standing above its interpolated neighbours, which is the interpolation and not a line. Tested among the measured channels alone over 2020, the 48 channels RAD1 measured hold 26 persistent lines, all below 500 kHz: eight below 100 kHz, where the solar wind's plasma line wanders, and 18 from 104 to 484 kHz in the range of the auroral kilometric radiation, with occupancies of 1.4 to 7.7 per cent and the day-to-day variation of that emission; the six measured channels above 700 kHz hold none.

STEREO. Table 1 lists the STEREO years. STEREO-A holds 7, 8 and 10 persistent lines in

Table 1: The STEREO years. Persistent lines exceed one per cent occupancy on thirty days or more; left are those outside the known sources. Spacecraft lines are held five times as often as by Wind or the other STEREO spacecraft in the same year; the others lie in service bands or in the range of auroral kilometric radiation.

Spacecraft	Year	Persistent	Left	Spacecraft lines (kHz)
STEREO-A	2008	7	0	1375, 1675
STEREO-A	2012	8	0	725, 1275, 1475, 1675, 2075
STEREO-A	2020	10	0	725, 1275, 1475, 1675, 6925, 10425
STEREO-B	2010	6	0	6975, 10425, 10475
STEREO-B	2012	6	0	6975, 10425, 10475

2008, 2012 and 2020 and STEREO-B 6 in each of 2010 and 2012, every one in a service band, in the range of auroral kilometric radiation, or made on the spacecraft. The spacecraft lines include 1275, 1475, 1675 and 2075 kHz, which Wind holds less than a fifth as often, 725 kHz in STEREO-A in 2012, which STEREO-B holds 0.7 per cent of the time against STEREO-A's 25, and 10425 kHz, which STEREO-B holds 87 to 90 per cent of the time in 2010 and 2012 and STEREO-A 0.08, 0.4 and 4.7 per cent in 2008, 2012 and 2020, while Wind's occupancy there stays near 0.7 per cent: a line of the receiver design that grew in STEREO-A.

4 Related work

Low-frequency searches. Searches for narrowband technosignatures at the lowest frequencies observed from the ground, near 100 MHz, have used wide-field arrays such as the Murchison Widefield Array [1], whose channels of tens of kilohertz and sensitivity of tens of janskys over hours make them, like this one, searches for powerful broadband-channel emitters rather than for hertz-wide lines; we find no search below the ionospheric cutoff.

Spacecraft radio receivers. The receivers of Wind and STEREO were built for solar radio bursts, interplanetary plasma waves and Earth's radio emissions, and their calibration against the Galaxy's background [8] makes them sensitive monitors of the whole low-frequency sky. The Radio Astronomy Explorer first showed how much terrestrial noise leaks to space [7].

5 Conclusion

No persistent narrowband line below the ionospheric cutoff, from 20 kHz to 16 MHz, lies outside the known natural, terrestrial and spacecraft sources in 27 years of Wind spectra and five years of STEREO spectra. Terrestrial leakage betrays itself by following the broadcast bands from day to day, and lines made on a spacecraft by being missing from the others.

The search covers lines present in more than one per cent of the minutes on a month of days, five robust deviations above the neighbouring channels in a minute's average, which in RAD2 is $2 \times 10^{-16}\,\mathrm{W\,m^{-2}}$ in a 20 kHz channel: a transmitter of 6×10^{7} W radiating isotropically anywhere within 1 au, or of 2×10^{18} W at 1 pc, so the result bounds emitters in the Solar System and only the most powerful conceivable ones at the nearest stars, where a beam gains nothing at these wavelengths. A transmitter that is brief, drifting or broader than a channel is not constrained, nor is one inside the bands of terrestrial services, which are set aside whole and cover two thirds of the band above 1 MHz; RAD1 constrains only its 32 to 38 measured channels. Below a few hundred kilohertz the warm interstellar clouds absorb, so that part of the band bounds transmitters within

the heliosphere and along the shortest paths through warm gas. The STEREO spacecraft's direction finding would give the arrival direction of any line the spacecraft share, and the years of STEREO not yet searched would extend the comparison across the spacecraft's whole drift.

Acknowledgements

This work uses Wind/WAVES and STEREO/WAVES data from NASA's Coordinated Data Analysis Web (CDAWeb) at the Space Physics Data Facility, and the software NumPy, SciPy and cdflib.

References

[1] S. J. Tingay, C. Tremblay, A. Walsh and R. Urquhart, An opportunistic search for extraterrestrial intelligence (SETI) with the Murchison Widefield Array, *Astrophys. J. Lett.* **827**, L22 (2016).

[2] G. B. Rybicki and A. P. Lightman, *Radiative Processes in Astrophysics* (Wiley, New York, 1979).

[3] S. Redfield and J. L. Linsky, The structure of the local interstellar medium. IV. Dynamics, morphology, physical properties, and implications of cloud-cloud interactions, *Astrophys. J.* **673**, 283 (2008).

[4] J.-L. Bougeret, M. L. Kaiser, P. J. Kellogg, et al., WAVES: the radio and plasma wave investigation on the Wind spacecraft, *Space Sci. Rev.* **71**, 231 (1995).

[5] J.-L. Bougeret, K. Goetz, M. L. Kaiser, et al., S/WAVES: the radio and plasma wave investigation on the STEREO mission, *Space Sci. Rev.* **136**, 487 (2008).

[6] D. A. Gurnett, The Earth as a radio source: terrestrial kilometric radiation, *J. Geophys. Res.* **79**, 4227 (1974).

[7] J. R. Herman, J. A. Caruso and R. G. Stone, Radio Astronomy Explorer (RAE)-I observations of terrestrial radio noise, *Planet. Space Sci.* **21**, 443 (1973).

[8] G. A. Dulk, W. C. Erickson, R. Manning and J.-L. Bougeret, Calibration of low-frequency radio telescopes using the galactic background radiation, *Astron. Astrophys.* **365**, 294 (2001).

[9] H. V. Cane, Spectra of the non-thermal radio radiation from the galactic polar regions, *Mon. Not. R. Astron. Soc.* **189**, 465 (1979).